

Chapter 12

Mass-Storage Systems



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Overview of Storage

Primary Memory

- A processor or computer initially or directly accesses primary memory while using a computer.
- It enables a processor to access programs and services that are now in use and temporarily stored in a particular area of memory.
- The volatile storage component of a computer system is primary memory. Although there may be data buses, cache memory, or Random Access Memory (RAM), RAM is the most common example for primary memory.
- An operating system (OS), user interface, and any user-installed and running software utilities are all loaded into main memory as soon as a computer turns on.
- When a program or application is launched from main memory, it communicates with the system processor to carry out all of its unique functions.

Overview of Storage

Secondary Memory

- Secondary memory is non-volatile, permanent computer memory that is not directly accessible by a computer or processor.
- Data that can be quickly and easily retrieved, transmitted, and utilized by apps and services can be stored by the user and then used in this manner.
- Read-only memory (ROM), flash drives, hard disk drives (HDD), magnetic tapes, and other forms of internal and external storage media are all considered secondary memory. Only the major or main memory may access secondary memory during computations, and only then is it sent to the processor.
- Even if the computer is not powered on, data can be stored and retained in secondary memory, which is slower than primary memory. Additionally, it has large storage capacity, with each memory being capable of holding anywhere from a few megabytes (MB) to many terabytes (TB).
- Secondary memory is said to be slower than primary memory.
- All the mass storage media belongs to the secondary memory.

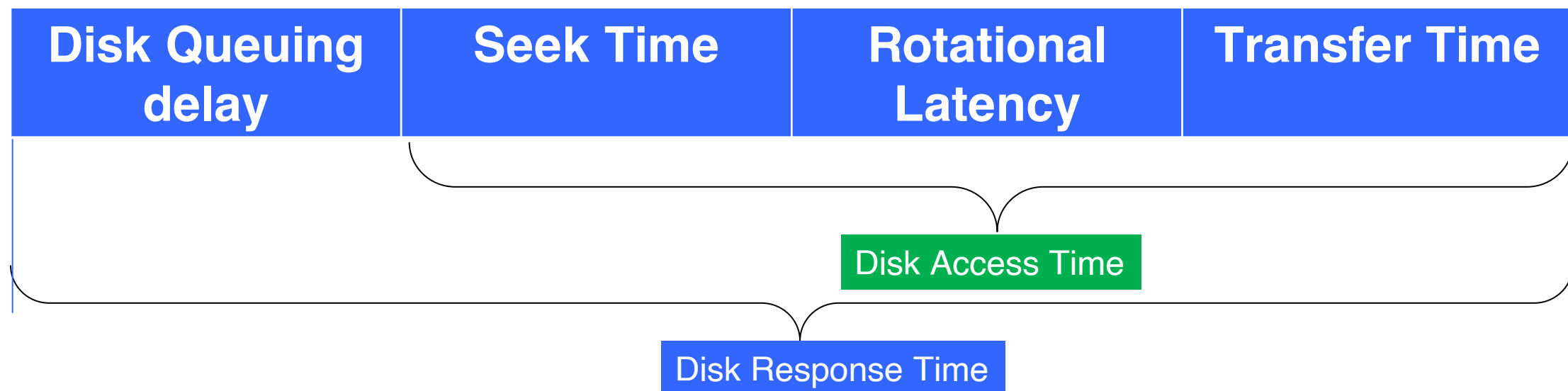
Disk Scheduling

- **Disk scheduling** is done by operating systems to schedule I/O requests arriving for the disk. **Disk scheduling is also known as I/O scheduling.**
- Disk scheduling is important because:
 1. Multiple I/O requests may arrive by different processes and only one I/O request can be served at a time by the disk controller. Thus other I/O requests need to wait in the waiting queue and need to be scheduled.
 2. Two or more request may be far from each other so can result in greater disk arm movement.
 3. Hard drives are one of the slowest parts of the computer system and thus need to be accessed in an efficient manner.

Disk Scheduling

Purpose of Disk Scheduling

The main purpose of disk scheduling algorithm is to select a disk request from the queue of IO requests and decide the schedule when this request will be processed.



■ Goal of Disk Scheduling Algorithm

- Fairness
- High throughput
- Minimal traveling head time

- **Seek Time** → Move head to correct track
- **Rotational Latency** → Wait for the required sector to rotate under the head
- **Transfer Time** → Actual reading/writing of data

Disk Scheduling (Cont.)

■ Disk Scheduling Algorithms

The list of various disks scheduling algorithm is given below. Each algorithm is carrying some advantages and disadvantages. The limitation of each algorithm leads to the evolution of a new algorithm.

1. FCFS scheduling algorithm
2. SSTF (shortest seek time first) algorithm
3. SCAN scheduling
4. C-SCAN scheduling
5. LOOK Scheduling
6. C-LOOK scheduling

FCFS

FCFS: FCFS is the simplest of all the Disk Scheduling Algorithms. In FCFS, the requests are addressed in the order they arrive in the disk queue.

Advantages:

- Every request gets a fair chance
- No indefinite postponement

Disadvantages:

- Does not try to optimize **seek time**
- May not provide the best possible service

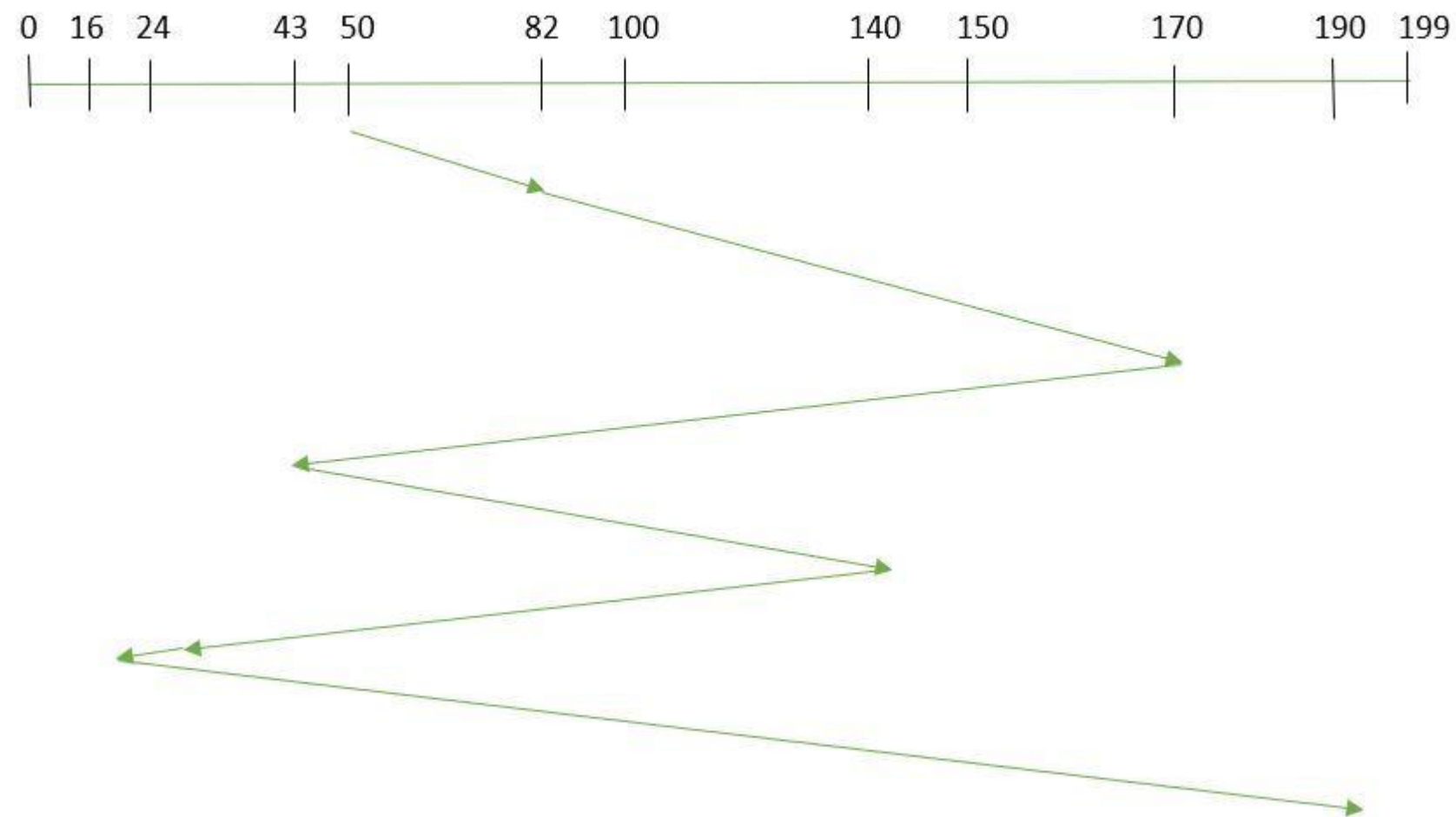
Seek Time is the time required for a disk drive's **read/write head** to move from its current track to the track where the requested data is stored. It is one of the major components of disk access time.

FCFS

Example:

1. Suppose the order of request is- (82,170,43,140,24,16,190)

And current position of Read/Write head is : 50



So, total seek time:

$$\begin{aligned} &= (82 - 50) + (170 - 82) + (170 - 43) + (140 - 43) + (140 - 24) + (24 - 16) + (190 - 16) \\ &= 642 \end{aligned}$$

SSTF

- **SSTF**: In SSTF (Shortest Seek Time First), requests having shortest seek time are executed first.
- So, the seek time of every request is calculated in advance in the queue and then they are scheduled according to their calculated seek time.
- As a result, the request near the disk arm will get executed first. SSTF is certainly an improvement over FCFS as it decreases the average response time and increases the throughput of system.

Advantages:

- Average Response Time decreases
- Throughput increases

Disadvantages:

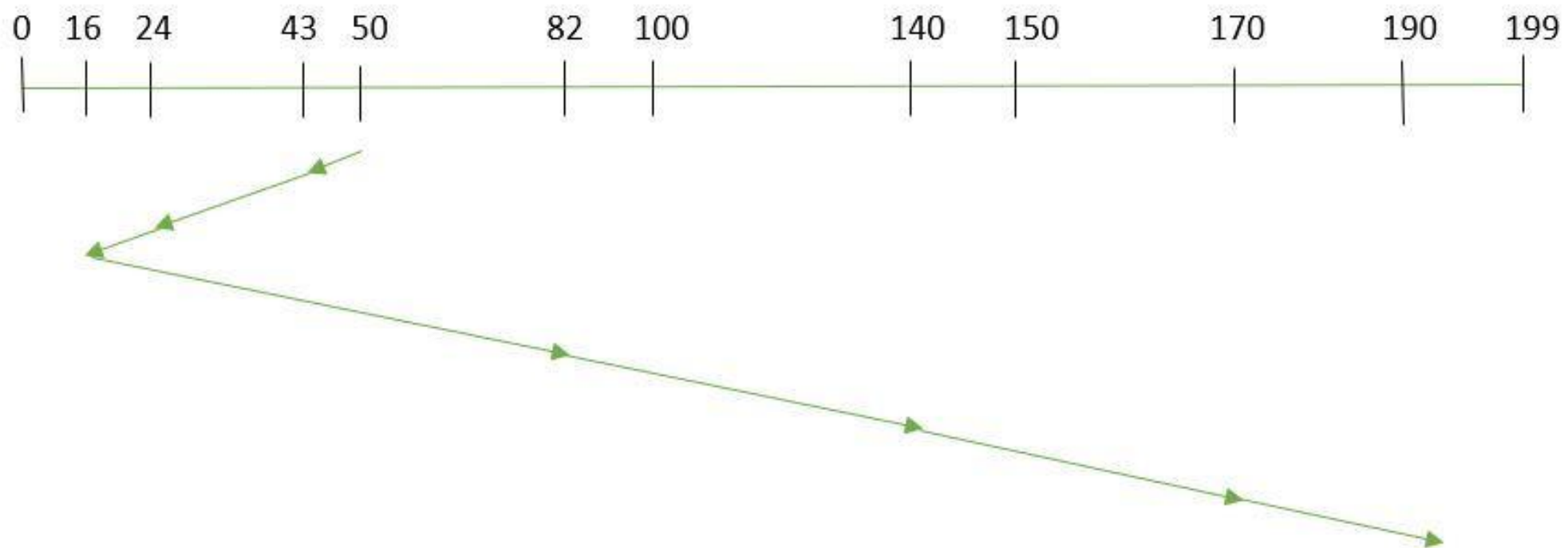
- Overhead to calculate seek time in advance
- Can cause Starvation for a request if it has higher seek time as compared to incoming requests
- High variance of response time as SSTF favors only some requests

SSTF (Cont.)

Example:

1. Suppose the order of request is- (82,170,43,140,24,16,190)

And current position of Read/Write head is : 50



So, total seek time:
$$=(50-43)+(43-24)+(24-16)+(82-16)+(140-82)+(170-140)+(190-170)$$
$$=208$$

SCAN

- In SCAN algorithm the disk arm moves into a particular direction and services the requests coming in its path and after reaching the end of disk, it reverses its direction and again services the request arriving in its path.
- So, this algorithm works as an elevator and hence also known as **elevator algorithm**. As a result, the requests at the midrange are serviced more and those arriving behind the disk arm will have to wait.

Advantages:

- High throughput
- Low variance of response time
- Average response time

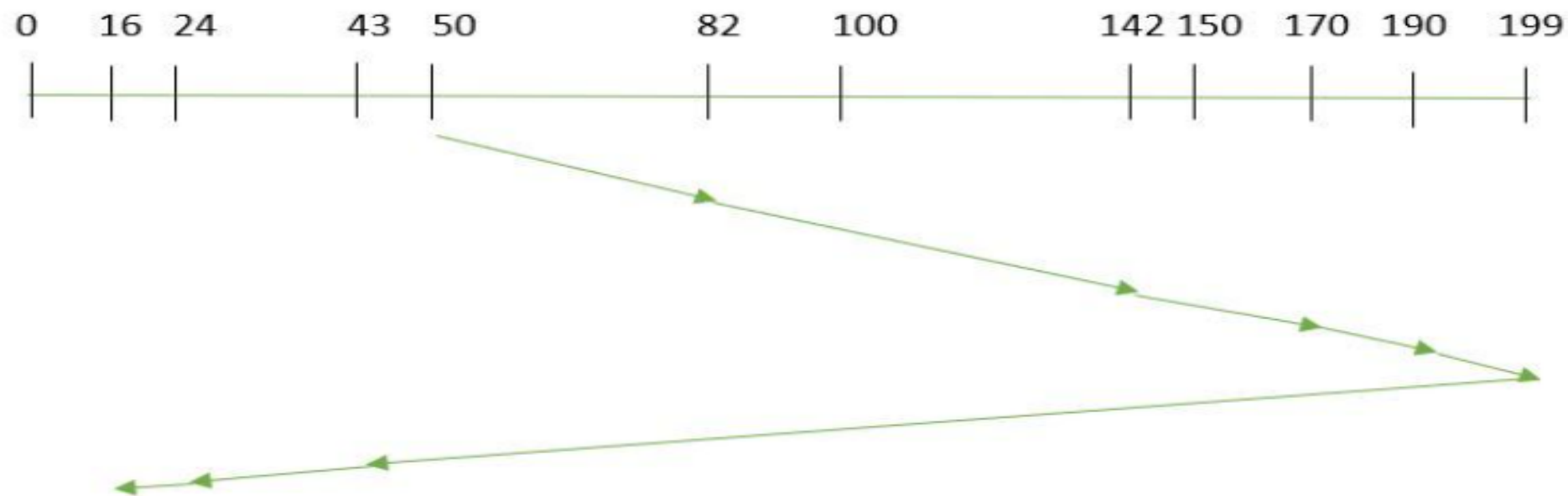
Disadvantages:

- Long waiting time for requests for locations just visited by disk arm

SCAN (Cont.)

Example:

1. Suppose the requests to be addressed are -82,170,43,140,24,16,190. And the Read/Write arm is at 50, and it is also given that the disk arm should move **"towards the larger value"**.



Therefore, the seek time is calculated as:

$$\begin{aligned} 1. &= (199 - 50) + (199 - 16) \\ &= 332 \end{aligned}$$

C-SCAN

- **CSCAN**: In SCAN algorithm, the disk arm again scans the path that has been scanned, after reversing its direction. So, it may be possible that too many requests are waiting at the other end or there may be zero or few requests pending at the scanned area.
- These situations are avoided in *CSCAN* algorithm in which the disk arm instead of reversing its direction goes to the other end of the disk and starts servicing the requests from there. So, the disk arm moves in a circular fashion and this algorithm is also similar to SCAN algorithm and hence it is known as C-SCAN (Circular SCAN).

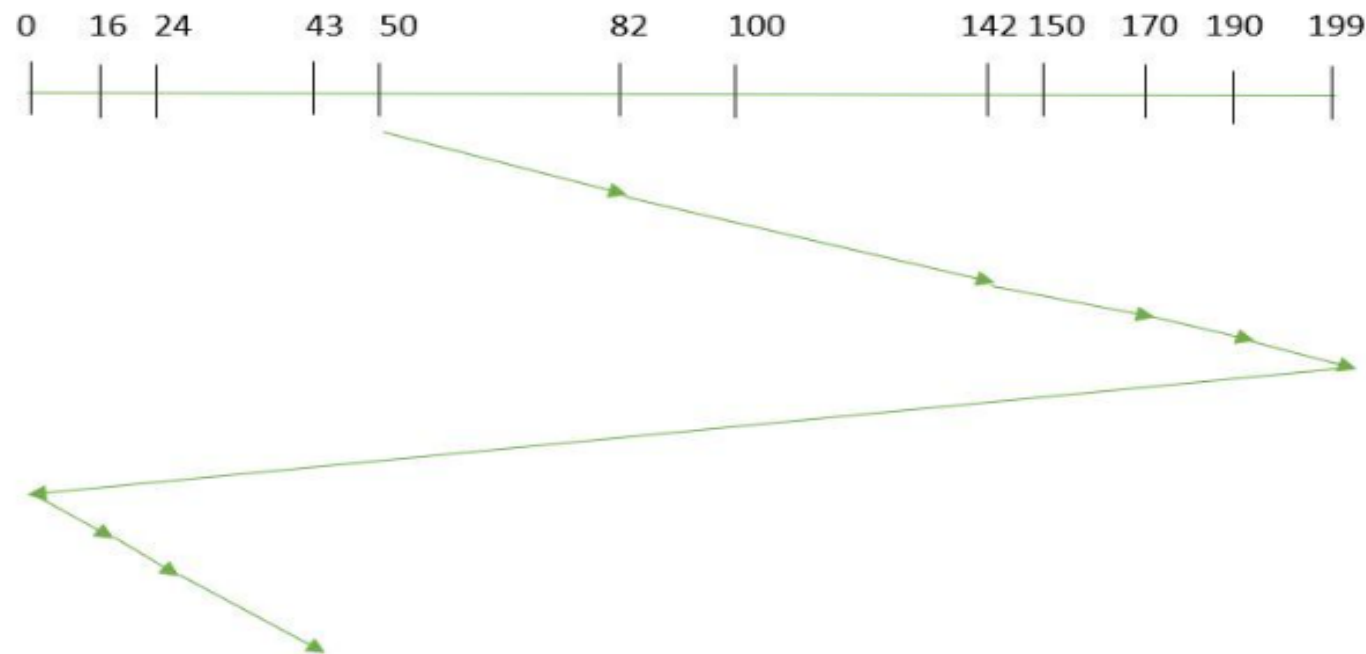
Advantages:

- Provides more uniform wait time compared to SCAN

C-SCAN (Cont.)

Example:

Suppose the requests to be addressed are -82,170,43,140,24,16,190. And the Read/Write arm is at 50, and it is also given that the disk arm should move **"towards the larger value"**.



Seek time is calculated as:

$$=(199-50)+(199-0)+(43-0)$$

$$=391$$

Look

- **LOOK:** It is similar to the SCAN disk scheduling algorithm except for the difference that the disk arm in spite of going to the end of the disk goes only to the last request to be serviced in front of the head and then reverses its direction from there only. Thus it prevents the extra delay which occurred due to unnecessary traversal to the end of the disk.

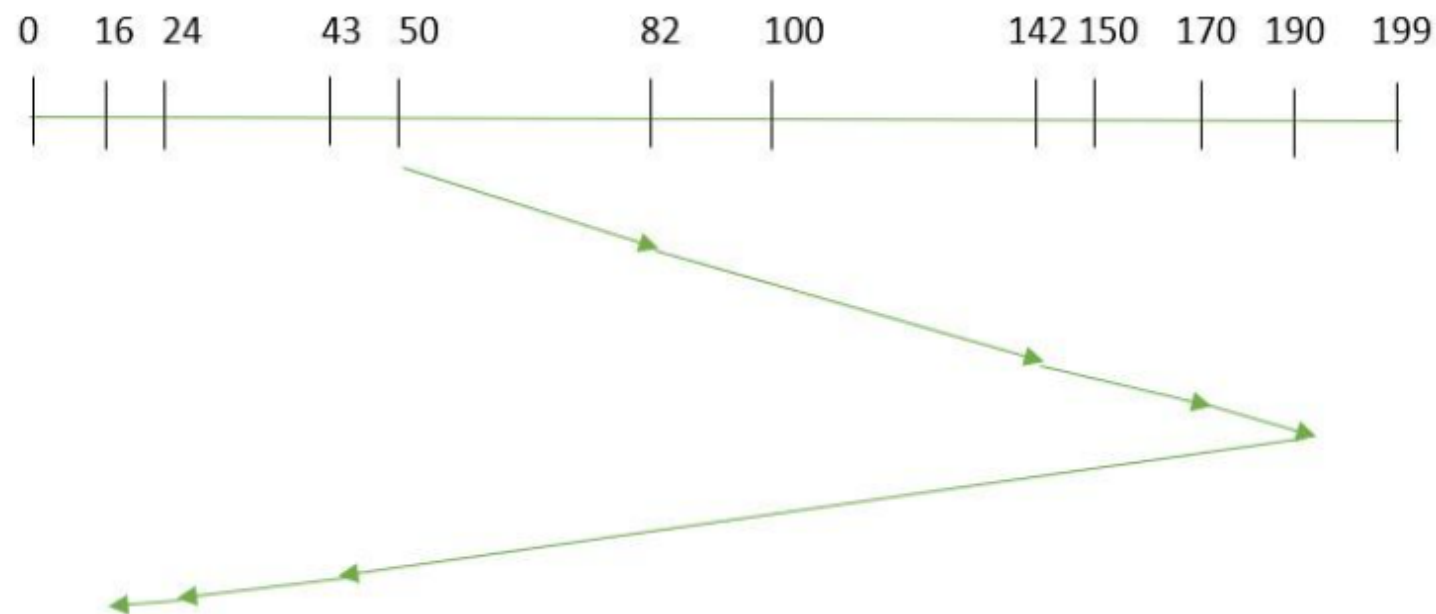
Advantages:

- Provides more uniform wait time compared to SCAN

LOOK (Cont.)

Example:

1. Suppose the requests to be addressed are -82,170,43,140,24,16,190. And the Read/Write arm is at 50, and it is also given that the disk arm should move **"towards the larger value"**.



So, the seek time is calculated as:

$$\begin{aligned} 1. &= (190 - 50) + (190 - 16) \\ &= 314 \end{aligned}$$

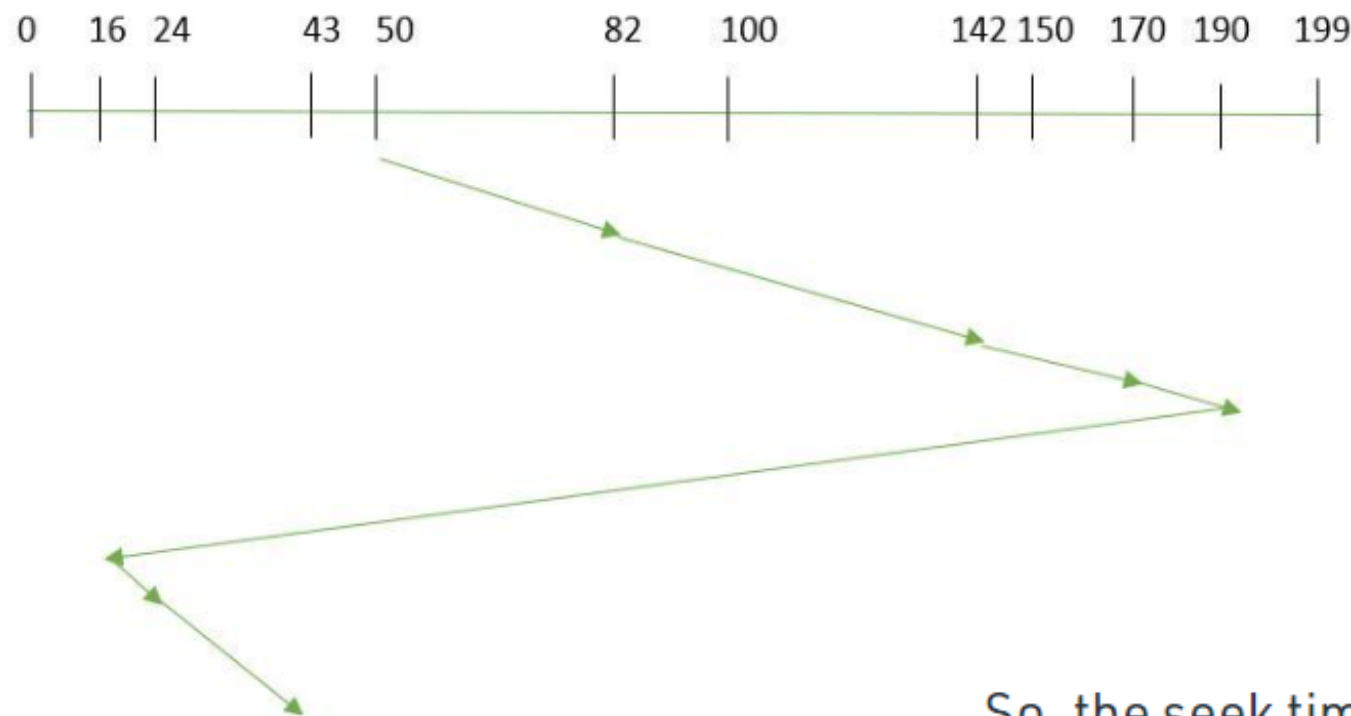
C-LOOK

- **CLOOK:** As LOOK is similar to SCAN algorithm, in similar way, CLOOK is similar to CSCAN disk scheduling algorithm. In CLOOK, the disk arm in spite of going to the end goes only to the last request to be serviced in front of the head and then from there goes to the other end's last request. Thus, it also prevents the extra delay which occurred due to unnecessary traversal to the end of the disk.

C-LOOK (Cont.)

Example:

1. Suppose the requests to be addressed are -82,170,43,140,24,16,190. And the Read/Write arm is at 50, and it is also given that the disk arm should move **"towards the larger value"**



So, the seek time is calculated as:

$$\begin{aligned} 1. &= (190 - 50) + (190 - 16) + (43 - 16) \\ &= 341 \end{aligned}$$

Selecting a Disk-Scheduling Algorithm

- SSTF is common and has a natural appeal
- SCAN and C-SCAN perform better for systems that place a heavy load on the disk
 - Less starvation
- Performance depends on the number and types of requests
- The disk-scheduling algorithm should be written as a separate module of the operating system, allowing it to be replaced with a different algorithm if necessary
- Either SSTF or LOOK is a reasonable choice for the default algorithm

Overview of Mass Storage Structure

- Systems designed to store enormous volumes of data are referred to as mass storage devices.
- Massive storage devices are sometimes used interchangeably with peripheral storage, which is the management of bigger volumes of data that are larger than the native storage capability of a computer or device.
- The basic idea of Mass Storage is to create a Data Backup or Data Recovery System.
- Today, mass storage may include several kinds of hard disks or solid-state storage devices, as well as tape drives and other physical data storage devices.
- The Mass Storage Structure Devices are:
 - Magnetic Disks
 - Solid State Disks
 - Magnetic Tapes

Magnetic Disks

- The process of magnetization is used to write, rewrite, and access data on a magnetic disk, a storage device. This process is known as Magnetic Disk. It is coated magnetically and has tracks, spots, and sectors for storing data.
- In 1956, IBM created the first magnetic hard drive, a substantial device. Despite being large, it could only hold 5 megabytes of information. Since then, magnetic disks' storage capabilities have multiplied dramatically while simultaneously shrinking in size.
- Basic Common Examples of Magnetic Disks are:
 - Floppy Disks
 - Hard Disks
 - Zip Disks



Magnetic Disks

Structure and Working of Magnetic Disks:

- The basic structure of Magnetic Disks is:
- A mechanical arm that travels across a revolving magnetic surface, known as the platter, makes up the majority of a magnetic disk. They come together to make a "comb." Both reading from and writing to the disk are done using the mechanical arm. A magnetization process is used to read and write data on magnetic disks.
- One or more disk-shaped platters with magnetic material covering them. Unlike "floppy" disks, which are composed of more flexible plastic, hard disk platters are built of stiff metal.
- There are two work areas on each plate. The very top and bottom surfaces of a stack of platters were occasionally avoided by older hard disk drives because they are more prone to damage or even breaking in some cases.
- The arm's head slides across the platter's surface while it continues to spin rapidly. The head floats on a small layer of air since the entire apparatus is hermetically sealed. Tiny patches on the disk surface are magnetized and data is saved when a small current is delivered to the head. When the head needs to read the data, a little current could be delivered to such tiny locations on the platter.

Magnetic Disks

- On the disk, data is arranged in sectors and tracks, with tracks serving as the disk's spherical divisions. Blocks of data are separated into sectors, which are a subset of tracks. The sectors are where the magnetic disk's read and write operations are carried out. Because the tracks are so close together, the floating heads need to be controlled extremely precisely while reading or writing data.
- The tracks are a series of concentric circles that separate each working surface. A cylinder is a group of tracks that are all at the same distance from the platter's edge, or in the accompanying figure, all the tracks that are directly above one another.
- A piece track is further split into sectors, generally storing 512 bytes of data each, however some newer drives occasionally employ greater sector sizes.
- (Sectors also include a header and a trailer, which among other things contain checksum data. Although internal fragmentation and the quantity of disk that has to be designated bad in the event of mistakes are increased by larger sector sizes, the percentage of the disk used by headers and trailers is reduced.)

Magnetic Disks

- Read-write heads are used to read data from hard drives. In the typical setup (shown below), one head is used for each surface, mounted on a separate arm, and moved concurrently from one cylinder to the next by a shared arm assembly. (Other designs, such as separate read-write heads, might accelerate disk access but present significant technical challenges.)
- The number of heads (or working surfaces) times the number of tracks per surface times the number of sectors per track times the number of bytes per sector equals the storage capacity of a conventional disk drive. The head-sector-cylinder number of a specific physical block of data can be used to identify it.
- The time needed to move the heads from one cylinder to another and for the heads to settle down after the transfer is known as the positioning time, also known as the seek time or random access time. This is usually the stage that moves slowly and is the main obstacle to high transfer rates.
- The time it takes for the requested sector to spin and enter the read-write head is known as the rotational latency.

Magnetic Disks

- This can be anything from 0 and 1 complete revolutions, with an average of 1/2 revolutions. This is a physical action that often follows seek time as the second-slowest step. (If a disk rotates at 7200 revolutions per minute, the average rotational delay is $1/2$ revolution / 120 revolutions per second, or just over 4 milliseconds, a long time by computer standards.)
- The amount of time needed to electronically transmit data from a disk to a computer is called the transfer rate. (Some writers could also use the phrase "transfer rate" to refer to the entire transfer rate, which includes search time, rotational delay, and the transfer rate of electronic data.)
- Over the surface, disk heads "fly" on a paper-thin air cushion. In the event that they should unintentionally make contact with the disk, a head crash takes place, which may or may not permanently harm or even entirely destroy the disk. Due of this, it is customary to park the disk heads while shutting off a computer, which entails moving the heads off the disk or to an empty space on the disk.
- Typically, floppy disks are removeable. Hard drives may also be taken out and replaced with fresh ones, and some of them can even be swapped out while the computer is still functioning.

Magnetic Disks

- The I/O Bus, a connection used to connect disk drives to computers, is used for this purpose. Enhanced Integrated Drive Electronics (EIDE), Advanced Technology Attachment (ATA), Serial ATA (SATA), Universal Serial Bus (USB), Fiber Channel (FC), and Small Computer Systems Interface (SCSI) are a few of the popular interface types.
- The disk controller is included within the disk itself, whereas the host controller is located at the I/O bus's computer end. Via Input Output ports, the CPU sends orders to the host controller. The disk controller moves information from the onboard cache to the host controller and motherboard memory at electronic speeds after moving information between the magnetic surface and onboard cache.

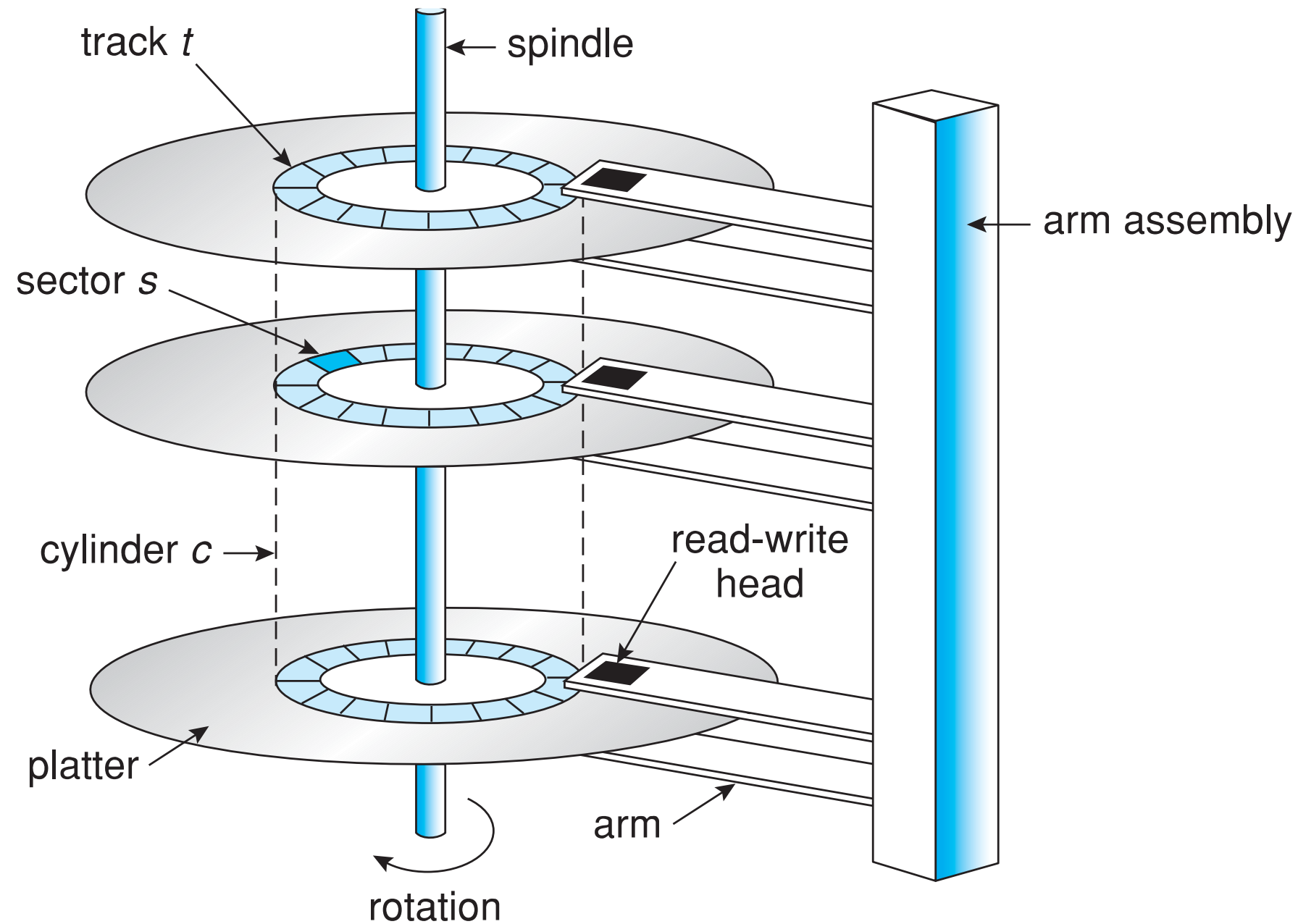
Magnetic Disks

- Platters range from .85” to 14” (historically)
 - Commonly 3.5”, 2.5”, and 1.8”
- Range from 30GB to 3TB per drive
- Performance
 - Transfer Rate – theoretical – 6 Gb/sec
 - Effective Transfer Rate – real – 1Gb/sec
 - Seek time from 3ms to 12ms – 9ms common for desktop drives
 - Average seek time measured or calculated based on 1/3 of tracks
 - Latency based on spindle speed
 - ▶ $1 / (\text{RPM} / 60) = 60 / \text{RPM}$
 - Average latency = $\frac{1}{2}$ latency

Spindle [rpm]	Average latency [ms]
4200	7.14
5400	5.56
7200	4.17
10000	3
15000	2

(From Wikipedia)

Moving-head Disk Mechanism



Disk Structure

- Disk drives are addressed as large 1-dimensional arrays of **logical blocks**, where the logical block is the smallest unit of transfer
 - Low-level formatting creates **logical blocks** on physical media
- The 1-dimensional array of logical blocks is mapped into the sectors of the disk sequentially
 - Sector 0 is the first sector of the first track on the outermost cylinder
 - Mapping proceeds in order through that track, then the rest of the tracks in that cylinder, and then through the rest of the cylinders from outermost to innermost
 - Logical to physical address should be easy
 - ▶ Except for bad sectors
 - ▶ Non-constant # of sectors per track via constant angular velocity

Magnetic Disk Performance

Some important terms related to Magnetic Disks:

- **Seek Time:** Seek time is the time taken in locating the disk arm to a specified track where the read/write request will be satisfied.
- **Rotational Latency:** It is the time taken by the desired sector to rotate itself to the position from where it can access the R/W heads.
- **Transfer Time:** It is the time taken to transfer the data.
- **Disk Access Time:** Disk access time is given as,
 - Disk Access Time = Rotational Latency + Seek Time + Transfer Time
- **Disk Response Time:** It is the average of time spent by each request waiting for the IO operation.

Magnetic Tapes

- Prior to the advent of hard disk drives, magnetic tapes were frequently utilized for secondary storage; today, they are mostly used for backups.
- It might take a while to get to a specific location on a magnetic tape, but once reading or writing starts, access rates are on par with disk drives.
- Tape drive capacities may be anywhere from 20 and 200 GB, and compression can increase that capacity by double.

Solid State Disks

- Old technologies are frequently employed in new ways as economic conditions and technology evolve. The growing usage of solid state drives, or SSDs, is one illustration of this.
- SSDs function as a tiny, quick hard disk using memory technology. To maintain the information over power cycles, certain implementations may employ either flash memory or DRAM chips protected by a battery.
- Due to the lack of moving components, SSDs operate far more quickly than conventional hard drives, and some issues, such the scheduling of disk accesses, simply do not exist.
- As a high-speed cache for information on hard disks that has to be retrieved fast, SSDs are particularly helpful. One use is to store frequently requested file system meta-data, such as directory and I node information. A boot drive is another version that has the OS and certain application executables but no essential user data. In order to make laptops thinner, lighter, and quicker, SSDs are also employed in them.
- The throughput of the bus may become a limiting problem due to how much quicker SSDs are than conventional hard drives, which leads to certain SSDs being linked directly to the system PCI bus.

End

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